

Electrical Circuit Formula

For preparation formula

1 DC circuit Analysis

Current(I): Rate of change flow.

$$I = \frac{dq}{dt}$$

voltage(V): Energy per unit charge. $V = \frac{dw}{dq}$, where dw = the amount of work done.

Power(P): Rate of energy absorption/supply.

$$P = VI = I^2R = \frac{V^2}{R}$$

Energy(W): Integrated power over time.

$$W = \int P dt$$

Ohm's Law:

$$V = IR$$

Kirchhoff's Current Law(KCL) and (KVL)

$$\sum I_{in} = \sum I_{out}$$

$$\sum V_{loop} = 0$$

Series and parallel circuits:

Series Resistance:

$$R_{eq} = R_1 + R_2 + \cdots + R_n$$

Parallel Resistance:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n} \quad (1)$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} \quad (2)$$

Voltage Divider (for R_1 in series with R_2):

$$V_1 = V_{total} * \left(\frac{R_1}{R_1 + R_2} \right)$$

Current Divider (for R_1 parallel to R_2):

$$I_1 = I_{total} * \left(\frac{R_2}{R_1 + R_2} \right)$$

Delta-Wye(Δ -Y) Conversions

Delta to Wye:

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} \quad (3)$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c} \quad (4)$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} \quad (5)$$

Wye to Delta:

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} \quad (6)$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} \quad (7)$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} \quad (8)$$

Circuit Network Theorems:

- Source Transformation: $V_s = I_s * R$ and $I_s = \frac{V_s}{R}$
- Thevenin Voltage and Resistance: $V_{Th} = V_{OC}$; (oc = Open circuit)

$$R_{Th} = \frac{V_{oc}}{I_{sc}} ; \text{ (Short-circuit current } I_{sc} \text{)}$$

- Norton Current And Resistance:

$$I_N = I_{sc}, R_N = R_{Th}$$

Inductors and Capacitors Capacitor(C):

- Current: $i(t) = C \frac{dv(t)}{dt}$
- Voltage: $V(t) = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + v(t_0)$
- Stored Energy: $W_c = \frac{1}{2} C V^2$
- DC steady State: Acts as an open circuit.

Inductor(L):

- Voltage: $v(t) = L \frac{di(t)}{dt}$
- Current: $i(t) = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$
- Stored Energy: $W_L = \frac{1}{2} L I^2$
- DC Steady State: Acts as a short circuit.

* First-order Transient Response (RC and RL Circuits):

$$x(t) = x(\infty) + [x(t_0) - x(\infty)] e^{-\frac{t}{\tau}}$$

For RC circuits: Time constant $\tau = R \cdot C$ For RL circuits: Time constant $\tau = \frac{L}{R}$

2 AC Sinusoidal Steady-State Analysis

1. Sinusoidal Basics and Waveform factors

- Sinusoidal Expression: $v(t) = V_m \sin(\omega t + \phi)$
- Angular Frequency: $\omega = 2\pi f = \frac{2\pi}{T}$
- Root-Mean-Square(RMS): $V_{rms} = \frac{V_m}{\sqrt{2}} \approx 0.707V_m$
- Average Value(Half-Cycle for sine wave): $V_{avg} = \frac{2V_m}{\pi} \approx 0.637V_m$
- Form Factor: Form Factor = $\frac{V_{rms}}{V_{avg}} = \frac{\pi}{2\sqrt{2}}$
- Peak Factor: Peak Factor = $\frac{V_m}{V_{rms}} = \sqrt{2} \approx 1.414$

2. Phasors, Impedance(Z), and Admittance(Y)

- Phasor Transform: $V(t) = V_m \cos(\omega t + \theta)$ and $V = V_m \angle \theta = V_m e^{j\theta}$
- Impedance($Z = R + jX$):
 - Resistor(R): $Z_R = R$
 - Inductor(L): $Z_L = j\omega L = jX_L$ (*Inductive Reactance* $X_L = \omega L$)
 - Capacitor(C): $Z_C = \frac{1}{j\omega C} = -jX_C$
Capacitance Reactance, $X_C = \frac{1}{\omega C}$
- Admittance ($Y = G + jB = \frac{1}{Z}$):
 - G. Conductance, B. Susceptance.

$$Y = \frac{1}{R+jX} = \frac{R}{R^2+X^2} - j \frac{X}{R^2+X^2}$$

3. AC Circuit Combination(R-L, R-C, R-L-C)

- Series RLC Impedance:

$$Z = R + j(\omega L - \frac{1}{\omega C}) = |Z| \angle \theta \quad |Z| = \sqrt{R^2 + (X_L - X_C)^2}, \quad \theta = \tan^{-1}(\frac{X_L - X_C}{R})$$

- AC Ohm's Law: $V = IZ$

4. Sinusoidal Steady-State Power

- Instantaneous Power: $p(t) = v(t) \cdot i(t)$
- Complex Power(S): $S = V_{rms} I_{rms} = P + jQ = |S| \angle \theta$
- Real/Average Power(P): Measured in Watts(W).

$$P = V_{rms} I_{rms} \cos \theta = I_{rms}^2 R$$

- Reactive Power(Q): Measured in Volt-Amperes Reactive(VAR)

$$Q = V_{rms} I_{rms} \sin \theta = I_{rms}^2 X$$

- Apparent Power(S): Measured in Volt-Ampere(VA).

$$S = |S| = V_{rms} I_{rms} = \sqrt{P^2 + Q^2}$$

- Power Factor(PF): $PF = \cos \theta = \frac{P}{S}$

5. AC Maximum Power Transfer and Impedance Matching

- For a load impedance $Z_L = Z_{Th} \Rightarrow R_L = R_{Th}$ and $X_L = -X_{Th}$
- Maximum Transferred Real Power:

$$P_{max} = \frac{|V_{Th,rms}|^2}{4R_{Th}}$$

3 Technique of General Circuit Analysis

- **Advanced Network Analysis Equations**

Node-Voltage Method(KCL Matrix):

When solving a circuit with n nodes using Nodal Analysis, you set a system of linear equations using admittance ($G = \frac{1}{R}$)

$$\begin{bmatrix} G_{11} & G_{12} & \dots \\ G_{21} & G_{22} & \dots \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

- Mesh-Current Method(KVL Matrix)

When solving a circuit with m meshes using Mesh Analysis, you set up a system using resistance(R):

$$\begin{bmatrix} R_{11} & R_{12} & \dots \\ R_{21} & R_{22} & \dots \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

- Superposition Theorem

For any linear variable x(voltage or current) affected by independent sources $S_1, S_2, \dots, S_n$:

$$x = x_1 + x_2 + \dots + x_n$$

(Where x_i is the value computed while keeping source S_i active and killing all other independent sources: short-circuiting voltage sources and open-circuiting current sources).

4 Component Combinations (Inductors and Capacitors)

- Inductors(L)

$$\begin{aligned} \text{Series: } L_{eq} &= L_1 + L_2 + \dots + L_n \\ \text{Parallel: } \frac{1}{L_{eq}} &= \frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_n} \end{aligned}$$

- Capacitance(C)

$$\begin{aligned} \text{Series: } \frac{1}{C_{eq}} &= \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n} \\ \text{Parallel: } C_{eq} &= C_1 + C_2 + \dots + C_n \end{aligned}$$

- RMS Value(Any Periodic Wave):

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$

- Average Value(Any periodic Wave):

$$V_{avg} = \frac{1}{T} \int_0^T v(t) dt$$

- Complex of AC Circuits:

Rectangular to Polar Conversion: $|Z| = \sqrt{R^2 + X^2}$, $\theta = \tan^{-1}(\frac{X}{R})$

- Polar to Rectangular Conversion:

$$R = |Z| \cos \theta, \quad X = |Z| \sin \theta$$

- Admittance Components Explicitly:

If $Z = R + jX$, then $Y = \frac{1}{Z} = G + jB$

$$\begin{aligned} G &= \frac{R}{R^2 + X^2} \Rightarrow (\text{Conductance}) \\ B &= \frac{-X}{R^2 + X^2} \Rightarrow (\text{Susceptance}) \end{aligned}$$

5 AC Voltage, Current and Power(R, L, C, R-L, R-C, R-L-C)

- Purely Resistive(R) Circuit:
 - Impedance: $Z = R \angle 0$
 - Phase: Voltage and Current are in phase($\theta = 0$)
 - Power Factor: $PF = \cos(0) = 1$
- Purely Inductive(L) Circuit:
 - Impedance: $Z = \omega L \angle 90 = j\omega L$
 - Phase: Current lags voltage by 90 ($0 = 90$).
 - Power Factor: $PF = \cos(90) = 0$ (lagging)
- Purely Capacitive(C) Circuit:
 - Impedance : $Z = \frac{1}{\omega C} \angle -90 = -j\frac{1}{\omega C}$
 - Phase: Current leads voltage by 90($\theta = -90$)
 - Power Factor: $PF = \cos(-90) = 0$ (leading)
- Series R-L Circuit:
 - Impedance: $Z = R + j\omega L = \sqrt{R^2 + (\omega L)^2} \angle \tan^{-1}(\frac{\omega L}{R})$
 - Current lags voltage by angle θ
- Series R-C Circuit:
 - Impedance: $Z = R - j\frac{1}{\omega C} = \sqrt{R^2 + (\frac{1}{\omega C})^2} \angle -\tan^{-1}(\frac{1}{\omega CR})$

6 AC Series-Parallel Simplification(R, L, C)

- Impedance in Series:

$$Z_{eq} = Z_1 + Z_2 + Z_3 + \dots$$
- Impedance in Parallel: $\frac{1}{Z_{eq}} = \frac{1}{Z_1} + \frac{1}{Z_2} + \dots$ or *for two component* : $Z_{eq} = \frac{Z_1 \cdot Z_2}{Z_1 + Z_2}$
- AC Delta-Wye ($\delta - Y$) Conversion
 The conversion formulas are structurally identical to the DC resistance formulas, but you use Complex Impedance(Z) instead of just R:
 - Delta (Δ) to Wye (Y):

$$Z_1 = \frac{Z_b Z_c}{Z_a + Z_b + Z_c},$$

$$Z_2 = \frac{Z_a Z_c}{Z_a + Z_b + Z_c},$$

$$Z_3 = \frac{Z_a Z_b}{Z_a + Z_b + Z_c}.$$

$$Z_a = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1},$$

$$Z_b = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2},$$

$$Z_c = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_3}.$$

- AC Maximum Power Transfer
 - * Reactive part X_L is fixed, and the magnitude $|Z_L|$ is varied.

*

$$R_L = \sqrt{R_{Th}^2 + (X_{Th} + X_L)^2}$$

* If the circuit is restricted to matching only the magnitudes,

$$|Z_L| = |Z_{Th}| = \sqrt{R_{Th}^2 + X_{Th}^2}$$